

Singular Value Decomposition

Theorem: Singular Value Theorem for Linear Transformation $\rightarrow$

Let V and W be finite-dimensional inner product spaces, and let $T: V \rightarrow W$ be a L.T of rank r . Then there exists orthonormal bases $\{v_1, v_2, \dots, v_n\}$ for V and $\{u_1, u_2, \dots, u_m\}$ for W and positive scalars $\sigma_1 > \sigma_2 > \dots > \sigma_r$ such that

$$T(v_i) = \begin{cases} \sigma_i u_i & \text{if } 1 \leq i \leq r \\ 0 & \text{if } i > r \end{cases}$$

Conversely, suppose that the preceding conditions are satisfied. Then for $1 \leq i \leq r$, u_i is an eigen vector of T^*T with corresponding eigen value σ_i^2 if $1 \leq i \leq r$ and 0 if $i > r$. Therefore the scalars $\sigma_1, \sigma_2, \dots, \sigma_r$ are uniquely determined by T .

Singular Values of Transformation

Defⁿ $\rightarrow$ The unique scalars $\sigma_1, \sigma_2, \dots, \sigma_r$ are called the singular values of T . If r is less than both m and n , then the term singular value is extended to include $\sigma_{r+1} = \dots = \sigma_k = 0$, where k is the minimum of m and n .

Singular Values of matrix

Defⁿ $\rightarrow$ Let A be a $m \times n$ matrix. we define the singular values of A to be the singular values of the linear transformation L_A .

Singular Value Decomposition Theorem for Matrices

Let A be an $m \times n$ matrix of rank r with the positive singular values $\sigma_1 > \sigma_2 > \dots > \sigma_r$, and let Σ be the $m \times n$ matrix defined by

$$\Sigma_{ij} = \begin{cases} \sigma_i & \text{if } i = j \leq r \\ 0 & \text{otherwise.} \end{cases}$$

Then there exists an $m \times m$ unitary matrix U and an $n \times n$ unitary matrix V such that $A = U \Sigma V^*$.

Singular Value Decomposition of A

Defⁿ Let A be an $m \times n$ matrix of rank r with positive singular value $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r$. A factorization $A = U \Sigma V^T$ where U and V are unitary matrices and Σ is the $m \times n$ matrix defined

$$\text{by } \Sigma_{ij} = \begin{cases} \sigma_i & \text{if } i=j \leq r \\ 0 & \text{otherwise} \end{cases}$$

is called a singular value decomposition of A .

Let A be an $m \times n$ matrix. Then $A = U \Sigma V^T$ is the singular value decomposition of A .

- U is $m \times m$ orthogonal matrix with columns equal to the unit eigenvectors of $A^T A$.

$$U = \begin{bmatrix} \vec{u}_1 & \vec{u}_2 & \vdots \end{bmatrix}$$

- V is an $n \times n$ orthogonal matrix whose columns are unit eigenvectors of $A^T A$.

$$A^T A V = \begin{bmatrix} \vec{v}_1 & \vec{v}_2 & \vdots \end{bmatrix}$$

- Σ is an $m \times n$ matrix with the singular values of A on the main diagonal and all other entries of zero.

$$\Sigma = \begin{bmatrix} \vec{\sigma}_1 & 0 & 0 \\ 0 & \vec{\sigma}_2 & \vdots \\ 0 & 0 & \vdots \end{bmatrix}$$

$$A_{2 \times 3} = U_{2 \times 2} \Sigma_{2 \times 3} V_{3 \times 3}^T$$

$$\left| \begin{array}{l} \text{Eigen values} \rightarrow |A - \lambda I| = 0 \\ \text{Eigen vectors} \rightarrow [A - \lambda I]x = 0 \end{array} \right.$$

Singular values of a matrix $\rightarrow$

Let A be $m \times n$ matrix

- The singular values of A are the square roots of the positive eigenvalues of $A^T A$ or $A A^T$.
- $A^T A$ and $A A^T$ have the same positive eigenvalues.

Steps $\rightarrow$

1. Determine V and then V^T
2. Determine the singular values σ_i and then Σ
3. Determine U using $A = U \Sigma V^T \rightarrow AV = U \Sigma$ since V is orthogonal to V^T , we know $V V^T = I$.

Q. $A = \begin{bmatrix} 1 & -1 & 3 \\ 3 & 1 & 1 \end{bmatrix}$ Find the Singular Values of A.

3.

$$A^T A = \begin{bmatrix} 1 & 3 \\ -1 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 3 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 2 & 6 \\ 2 & 2 & -2 \\ 6 & -2 & 10 \end{bmatrix}$$

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For eigen values, $|A - \lambda I| = 0$

$$\begin{vmatrix} 10-\lambda & 2 & 6 \\ 2 & 2-\lambda & -2 \\ 6 & -2 & 10-\lambda \end{vmatrix} = 0$$

ch. eqn. is $\lambda(\lambda-16)(\lambda-6) = 0 \Rightarrow \lambda_1 = 16, \lambda_2 = 6, \lambda_3 = 0$

(order from greatest to least)

Calc. eigenvectors for $\lambda_1 = 16$

$$[A - \lambda_1 I]X = 0$$

$$\begin{bmatrix} 10-16 & 2 & 6 \\ 2 & 2-16 & -2 \\ 6 & -2 & 10-16 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -6 & 2 & 6 \\ 2 & -14 & -2 \\ 6 & -2 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -6 & 2 & 6 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} x_1 - x_3 = 0 \\ x_2 = 0 \\ x_3 = x_3 \end{matrix} \Rightarrow \begin{matrix} x_1 = x_3 \\ x_2 = 0 \\ x_3 = t \end{matrix} \Rightarrow X_1 = \begin{bmatrix} t \\ 0 \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, t \neq 0$$

$$\vec{x}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

(normalizing the vector x_1)

For $\lambda_2 = 6$, $[A - \lambda_2 I]X = 0$

$$\begin{bmatrix} 10-6 & 2 & 6 \\ 2 & 2-6 & -2 \\ 6 & -2 & 10-6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 4 & 2 & 6 \\ 2 & -4 & -2 \\ 6 & -2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} x_1 + x_3 = 0 \\ x_2 + x_3 = 0 \\ x_3 = x_3 \end{matrix} \Rightarrow \begin{matrix} x_1 = -x_3 \\ x_2 = -x_3 \\ x_3 = t \end{matrix} \Rightarrow X_2 = \begin{bmatrix} -t \\ -t \\ t \end{bmatrix} = t \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, t \neq 0$$

$$\vec{x}_2 = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}$$

For $\lambda_3 = 0$, $[A - \lambda_3 I]X = 0$

$$\begin{bmatrix} 10 & 2 & 6 \\ 2 & 2 & -2 \\ 6 & -2 & 10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} x_1 + x_3 = 0 \\ x_2 - 2x_3 = 0 \\ x_3 = x_3 \end{matrix} \Rightarrow \begin{matrix} x_1 = -x_3 \\ x_2 = 2x_3 \\ x_3 = t \end{matrix} \Rightarrow X_3 = \begin{bmatrix} -t \\ 2t \\ t \end{bmatrix} = t \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, t \neq 0$$

$$\vec{x}_3 = \frac{1}{\sqrt{6}} \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}$$

Find SVD for

Q. $A = \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \\ 2 & 2 & -2 \end{pmatrix}$, $\lambda_1=6, \lambda_2=0, \lambda_3=0, \therefore \sigma_1=\sqrt{6}, \Sigma = \begin{bmatrix} \sqrt{6} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & -\frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix}$ and $V^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix}$

The singular values of AA^T in order from greatest to least are:

$\sigma_1 = \sqrt{\lambda_1} = \sqrt{6} = 2, \sigma_2 = \sqrt{\lambda_2} = \sqrt{6}$

$\Sigma = \begin{bmatrix} 4 & 0 & 0 \\ 0 & \sqrt{6} & 0 \end{bmatrix}$

$\Sigma = \text{singular values}$
 $= \sqrt{(\text{Eigenvalue})}$

Now we need to find U.

$A = U\Sigma V^T$

$AV = U\Sigma V^T V$

$AV = U\Sigma I$

Since V and V^T are orthogonal $V^T V = I$

$AV = U\Sigma$

It follows: $A\vec{v}_1 = \sigma_1 \vec{u}_1$

$A\vec{v}_2 = \sigma_2 \vec{u}_2$

$A\vec{v}_1 = \sigma_1 \vec{u}_1 \rightarrow \vec{u}_1 = \frac{1}{\sigma_1} A\vec{v}_1 = \frac{1}{4} \begin{bmatrix} 1 & -1 & 3 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$

$A\vec{v}_2 = \sigma_2 \vec{u}_2 \rightarrow \vec{u}_2 = \frac{1}{\sigma_2} A\vec{v}_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} 3 & -1 & 3 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} -\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$

$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$

~~Find~~ $A = U\Sigma V^T$

$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 4 & 0 & 0 \\ 0 & \sqrt{6} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix} = \begin{bmatrix} 1 & -1 & 3 \\ 3 & 1 & 1 \end{bmatrix}$

$U \Sigma V^T = A$

Q. Find SVD for the given matrix $A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \end{bmatrix}$.

Soln. $\rightarrow A^T A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & -2 \\ 2 & 2 & -2 \\ -2 & -2 & 2 \end{bmatrix}$

For eigenvalue $A - \lambda I = 0$

$$\begin{vmatrix} 2-\lambda & 2 & -2 \\ 2 & 2-\lambda & -2 \\ -2 & -2 & 2-\lambda \end{vmatrix} = 0 \quad \text{ch. eqn is}$$

$$(2-\lambda)[(2-\lambda)^2 - 4] - 2[2(2-\lambda) - 4] - 2[-4 + 2(2-\lambda)] = 0$$

$$\Rightarrow (2-\lambda)[4 - 4\lambda + \lambda^2 - 4] - 2[4 - 2\lambda - 4] - 2[-4 + 4 - 2\lambda] = 0$$

$$\Rightarrow (2-\lambda)(\lambda^2 - 4\lambda) + 4\lambda + 4\lambda = 0$$

$$\Rightarrow 2\lambda^2 - 8\lambda - \lambda^3 + 4\lambda^2 + 8\lambda = 0$$

$$\Rightarrow -\lambda^3 + 6\lambda^2 = 0$$

$$\Rightarrow \lambda^3 - 6\lambda^2 = 0$$

$$\Rightarrow \lambda^2(\lambda - 6) = 0$$

$$\Rightarrow \lambda = 0, 0, 6$$

$$\therefore \lambda_1 = 6, \lambda_2 = 0, \lambda_3 = 0$$

Corr. eigen vector for $\lambda_1 = 6$ is $[A - \lambda I]x = 0$

$$\Rightarrow \begin{bmatrix} -4 & 2 & -2 \\ 2 & -4 & -2 \\ -2 & -2 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \therefore x_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \text{ normalizing the vector } x_1, \text{ we get}$$

$$v_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

Eigen vector Corr. to $\lambda_2 = 0 = \lambda_3$

$$\begin{bmatrix} 2 & 2 & -2 \\ 2 & 2 & -2 \\ -2 & -2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \therefore x_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \text{ normalizing the vector } x_2$$

$$v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$x_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \text{ normalizing the vector } x_3, \text{ and } v_3 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$\therefore V = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \end{bmatrix}, \quad V^T = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} \end{bmatrix}$$

The singular values of $A^T A$ in order from greatest to least are: -
 $\sigma_1 = \sqrt{6}$ is the only non-zero singular value of A .

$$\therefore \Sigma = \begin{bmatrix} \sqrt{6} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore V$ and V^T are orthogonal $V^T V = I$.

$$AV = U\Sigma$$

It follows: $Au_1 = \sigma_1 u_1$

$$Au_2 = \sigma_2 u_2$$

$$Au_1 = \sigma_1 u_1 \Rightarrow u_1 = \frac{1}{\sigma_1} Au_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \end{bmatrix}$$

$$\therefore u_1 = \frac{1}{\sqrt{6 \times 3}} \begin{bmatrix} 3 \\ 3 \end{bmatrix}_{2 \times 1} = \frac{3}{3\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

next choose $u_2 = \frac{1}{\sigma_2} Au_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, a unit vector orthogonal to u_1 ,

to obtain orthonormal basis $U = \{u_1, u_2\}$ for $\mathbb{R}^2$ and set

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Then $A = U\Sigma V^T$ is the desired SVD.

$$\therefore \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{6} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} \end{bmatrix} = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \end{bmatrix}$$

Pseudo inverse of $A =$

$$A^+ = V\Sigma^+ U^T = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ -1 & -1 \end{bmatrix}$$

Polar Decomposition $\rightarrow$ For any square matrix A , there exists a unitary matrix W and a positive semidefinite matrix P such that $A = WP$, furthermore, if A is invertible, then the representation is unique.

The factorization of a square matrix A as WP where W is unitary and P is positive definite is called a polar decomposition of A .

Q. Find the Polar decomposition of $A = \begin{bmatrix} 11 & -5 \\ -2 & 10 \end{bmatrix}$

Soln. $\rightarrow A^T A = \begin{bmatrix} 11 & -2 \\ -5 & 10 \end{bmatrix} \begin{bmatrix} 11 & -5 \\ -2 & 10 \end{bmatrix} = \begin{bmatrix} 125 & -75 \\ -75 & 125 \end{bmatrix}$

For eigenvalue, $|A - \lambda I| = 0$

$$\begin{vmatrix} 125 - \lambda & -75 \\ -75 & 125 - \lambda \end{vmatrix} = 0 \Rightarrow (125 - \lambda)^2 - (75)^2 = 0$$

$$\Rightarrow \lambda^2 - 250\lambda + 15625 - 5625 = 0$$

$$\Rightarrow \lambda^2 - 250\lambda + 10000 = 0$$

$$\Rightarrow \lambda^2 - 200\lambda - 50\lambda + 10000 = 0$$

$$\Rightarrow \lambda(\lambda - 200) - 50(\lambda - 200) = 0$$

$$\Rightarrow (\lambda - 50)(\lambda - 200) = 0 \Rightarrow \lambda_1 = 200, \lambda_2 = 50$$

Corr. eigen vectors for $\lambda_1 = 200$ i.e., $[A - \lambda I]X = 0$

$$\begin{bmatrix} 125 - 200 & -75 \\ -75 & 125 - 200 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -75 & -75 \\ -75 & -75 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Cor. eigenvector for $\lambda_1 = 200$ is

$$x_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\therefore v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Cor. eigenvector for $\lambda_2 = 50$ is

$$x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\therefore v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\therefore V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \text{ and } V^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

The singular values of AA^T are $\sigma_1 = \sqrt{200} = 10\sqrt{2}$, $\sigma_2 = \sqrt{50} = 5\sqrt{2}$

$$\text{and } \Sigma = \begin{bmatrix} 10\sqrt{2} & 0 \\ 0 & 5\sqrt{2} \end{bmatrix}$$

Next, we find the columns u_1 and u_2 of U

$$u_1 = \frac{1}{\sigma_1} A v_1 = \frac{1}{5} \begin{pmatrix} 4 \\ -3 \end{pmatrix} \text{ and } u_2 = \frac{1}{5} \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$$\therefore U = \begin{bmatrix} \frac{4}{5} & \frac{3}{5} \\ -\frac{3}{5} & \frac{4}{5} \end{bmatrix}$$

$$\therefore \text{we have } W = U V^T = \begin{bmatrix} \frac{4}{5} & \frac{3}{5} \\ -\frac{3}{5} & \frac{4}{5} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{5\sqrt{2}} \begin{bmatrix} 7 & -1 \\ 1 & 7 \end{bmatrix}$$

$$\text{and } P = V \Sigma V^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 10\sqrt{2} & 0 \\ 0 & 5\sqrt{2} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} = \frac{5}{\sqrt{2}} \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}$$

9
The Pseudoinverse of a Matrix $\rightarrow$ (Generalization of Matrix inverse)

Let A be a $m \times n$ matrix. Then there exists a unique $n \times m$ matrix B such that $(L_A)^+ : F^m \rightarrow F^n$ is equal to the left multiplication transformation L_B . We call B the pseudo inverse of A and denote it by $B = A^+$. Thus $(L_A)^+ = L_{A^+}$.

Th. Let A be an $m \times n$ matrix of rank r with a singular value decomposition $A = U \Sigma V^T$ and non-zero singular values $\sigma_1 > \sigma_2 > \dots > \sigma_r$. Let Σ^+ be the $n \times m$ matrix defined by

$$\Sigma_{ij}^+ = \begin{cases} \frac{1}{\sigma_i} & \text{if } i=j \leq r \\ 0 & \text{otherwise} \end{cases}$$

Then $A^+ = V \Sigma^+ U^T$, and this is a SVD of A^+ .

Note: Σ^+ is pseudo inverse of Σ .

~~SVD to Pseudo inverse~~

~~$A^+ = \frac{1}{\det A} A^T$~~

~~$A = U \Sigma V^T \Rightarrow A^+ = V \Sigma^+ U^T$~~

Q. Find A^+ for the matrix ~~$A = \begin{bmatrix} 1 & -1 & 3 \\ 3 & 1 & 1 \end{bmatrix}$~~ $A = \begin{bmatrix} 1 & -1 & 3 \\ 3 & 1 & 1 \end{bmatrix}$

Soln. $\rightarrow A^T A = \begin{bmatrix} 1 & 3 \\ -1 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 3 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 2 & 6 \\ 2 & 2 & -2 \\ 6 & -2 & 10 \end{bmatrix}$

Eigen values $|A - \lambda I| = 0$ are, $\lambda_1 = 16, \lambda_2 = 6, \lambda_3 = 0$

Corr. Eigen vectors are $X_1 = \begin{bmatrix} 1 \\ 6 \\ 1 \end{bmatrix}, X_2 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, X_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

$\therefore u_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}, u_2 = \begin{bmatrix} -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}, u_3 = \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}$

$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \\ 0 & -\frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix}, U^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix}$

Singular values of AA^T are $\sigma_1 = \sqrt{\lambda_1} = \sqrt{16} = 4, \sigma_2 = \sqrt{\lambda_2} = \sqrt{6}$

$\Sigma = \begin{bmatrix} 4 & 0 & 0 \\ 0 & \sqrt{6} & 0 \end{bmatrix}, U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$

$A = U \Sigma V^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 4 & 0 & 0 \\ 0 & \sqrt{6} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix}$

we have

$$A^+ = V \Sigma^+ U^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \\ 0 & -\frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} \frac{1}{4} & 0 & \text{~~0~~} \\ 0 & \frac{1}{\sqrt{6}} & \text{~~0~~} \\ 0 & 0 & \text{~~0~~} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= \frac{1}{96} \begin{bmatrix} 1 & 3 \\ -1 & 1 \\ 3 & 1 \end{bmatrix}$$

Low Rank Approximation $\rightarrow$

SVD provide a very simple solution to low rank approximation problem

Suppose $A \in \mathbb{R}^{m \times n}$, has SVD

$$A = U \Sigma V^T = \sum_{i=1}^n u_i \sigma_i v_i^T$$

then the k -approximation to A is given by

$$A_k = \sum_{i=1}^k u_i \sigma_i v_i^T \quad \text{where } k \leq \text{rank } A.$$

Let A be 5×5 matrix.

$$\sigma_1 = 3, \sigma_2 = 1, \sigma_3 = 0.5, \sigma_4 = 0.2, \sigma_5 = 0.05$$

$$A_{5 \times 5} = U_{5 \times 5} \begin{bmatrix} \sigma_1 & 0 & 0 & 0 & 0 \\ 0 & \sigma_2 & 0 & 0 & 0 \\ 0 & 0 & \sigma_3 & 0 & 0 \\ 0 & 0 & 0 & \sigma_4 & 0 \\ 0 & 0 & 0 & 0 & \sigma_5 \end{bmatrix} V_{5 \times 5}^T$$

$$\begin{aligned} A_3 &= U_{5 \times 3} \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix} V_{3 \times 5}^T \\ &= \begin{bmatrix} u_1 & u_2 & u_3 & u_4 & u_5 \\ | & | & | & | & | \end{bmatrix} \sum_{i=1}^3 \begin{bmatrix} \sigma_i & 0 & 0 & 0 & 0 \\ | & | & | & | & | \\ 0 & \sigma_i & 0 & 0 & 0 \\ | & | & | & | & | \\ 0 & 0 & \sigma_i & 0 & 0 \end{bmatrix} \end{aligned}$$

$$\underline{\underline{A_3}} = \begin{bmatrix} u_1 & u_2 & u_3 \\ | & | & | \end{bmatrix}_{3 \times 3} \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix}_{3 \times 3} \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}_{3 \times 3}^t$$

A measure of quality of the approximation is given by $\frac{\|A_K\|_F^2}{\|A\|_F^2} = \frac{\sigma_1^2 + \sigma_2^2 + \dots + \sigma_K^2}{\sigma_1^2 + \sigma_2^2 + \dots + \sigma_n^2}$

Ex Find rank 2 approximation of

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix} = U \Sigma V^t$$

$$A = \begin{bmatrix} 0.91 & 0.42 & 0.02 \\ 0.41 & -0.87 & -0.26 \\ 0.09 & -0.27 & 0.97 \end{bmatrix} \begin{bmatrix} 4.04 & 0 & 0 \\ 0 & 1.20 & 0 \\ 0 & 0 & 0.87 \end{bmatrix} \begin{bmatrix} 0.67 & 0.73 & 0.08 \\ 0.65 & -0.53 & -0.53 \\ 0.35 & -0.41 & 0.84 \end{bmatrix}^t$$

$$A_2 = \begin{bmatrix} 0.91 & 0.42 \\ 0.41 & -0.87 \\ 0.09 & -0.27 \end{bmatrix} \begin{bmatrix} 4.04 & 0 \\ 0 & 1.20 \end{bmatrix} \begin{bmatrix} 0.67 & 0.73 \\ 0.65 & -0.53 \\ 0.35 & -0.41 \end{bmatrix}^t$$

$$= \begin{bmatrix} 2.99 & 2.01 & 0.98 \\ 0.02 & 1.88 & 1.1 \\ -0.07 & 0.45 & 0.49 \end{bmatrix} \rightarrow$$